CS341 · COMPUTER GRAPHICS

Question Bank الحل

تدريب على كل شابترات الميد

محارب

01 حل A: صح وخطأ

#	الجواب	السبب
1	**True**	الـ display list = refresh display file.
2	**False**	هذا من عيوبه.
3	**True**	هو والـ laser light show.
4	**True**	تقنية التلفزيون.
5	**True**	dot matrix و laser.
6	**True**	0، 2، 4 ثم 1، 3، 5.
7	**False**	307,200 bytes = 38,400 **bits**.
8	**True**	
9	**False**	القليلة لـ animation، والعالية للصور الثابتة المعقدة.
10	**False**	LCD = nonemissive.
11	**False**	هذا الـ active. الـ passive: ترانزستور لكل صف أو عمود.
12	**True**	
13	**True**	
14	**True**	`0.3 + 0.7 = 1`
15	**False**	3D بس.
16	**False**	العكس: كل convex هو affine.

17	**False**	RGB جمعي، CMY طرحي.
18	**True**	الورق أبيض.
19	**True**	
20	**False**	DDA يستخدم كسور ثم يقرّب.

02 حل B: اختيار من متعدد

#	الجواب	السبب
1	**b**	vector, stroke-writing, calligraphic
2	**b**	الـ frame buffer للـ raster.
3	**c**	
4	**c**	Joystick جهاز إدخال.
5	**a**	
6	**c**	
7	**a**	
8	**b**	`√(9 + 16) = 5`
9	**b**	`a⊥ = (−ay, ax) = (−5, 2)`
10	**a**	والنقطة 1.
11	**c**	
12	**c**	
13	**b**	
14	**b**	
15	**c**	

03 حل C: الفراغات

1. scan line 2. pixel 3. refresh rate 4. dot pitch 5. complementary 6. shade
7. Magenta · Green 8. black (key) 9. hexcone · double cone 10. chromaticity
11. glu · glut 12. 2Δy – Δx

04 حل D: إجابات قصيرة

1. Random vs raster

Raster	Random	
Frame buffer (pixel values)	Display list (line commands)	Storage
Stair-step lines	Very high, smooth lines	Resolution
Easy	Limited, no realistic shading	Color / shading

2. Random scan

Advantages: very high resolution; easy animation; little memory.
Disadvantages: needs an intelligent, processor-controlled beam; flicker limits screen density; can't draw complex images; limited and expensive color.

3. CRT parts

Connector pins (base), electron gun, focusing system, vertical and horizontal deflection plates, phosphor-coated screen, electron beam.
The deflection plates steer the beam so it can hit any spot on the screen.

Beam-penetration vs shadow-mask .4

Beam-penetration: layers of colored phosphor; the color depends on how far the beam penetrates.
Shadow-mask: three electron guns (R, G, B) and a mask whose holes line up with triangles of red, green and blue phosphor dots.

Active vs passive matrix .5

Active: one transistor per pixel; brighter and sharper; more complicated and expensive.
Passive: one transistor per row or column; cheaper; needs less power.

Additive vs subtractive .6

Additive (RGB): lights are added; black is the origin; used by displays; R + G + B = white.
Subtractive (CMY): pigments absorb light; white is the origin; used by printers; C + M + Y = black.

CIE chromaticity diagram uses .7

Comparing color gamuts; identifying complementary colors; determining purity and dominant wavelength.

?Why K .8

Mixing C, M and Y does not produce a fully saturated black.